# Agora: continue or skip?

Does this architecture fit our control-plane study?

### Stage placement

Assigned centrally at entry;
workers stay in their assigned stage.

### Group formation

Rosters rebuilt each averaging round;
no Hivemind-style leader election.

### Hivemind lineage

Shared foundations, with different
coordination and trust assumptions.

### Possible research angle

Can a peer influence its stage or
trainer/worker assignment? Untested.

**Continue: assess Agora as a comparison case**

**Skip: save time for the main study**

Sources: Agora paper §§3.3.1, 3.6.1; pinned matchmaking implementation. Research question is untested.

# Pluralis Agora

## The DHT-based architecture behind the Pluralis-8B run

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# Pluralis-8B at a glance

**8B**
parameters

**7**
pipeline stages

**40 days**
of training

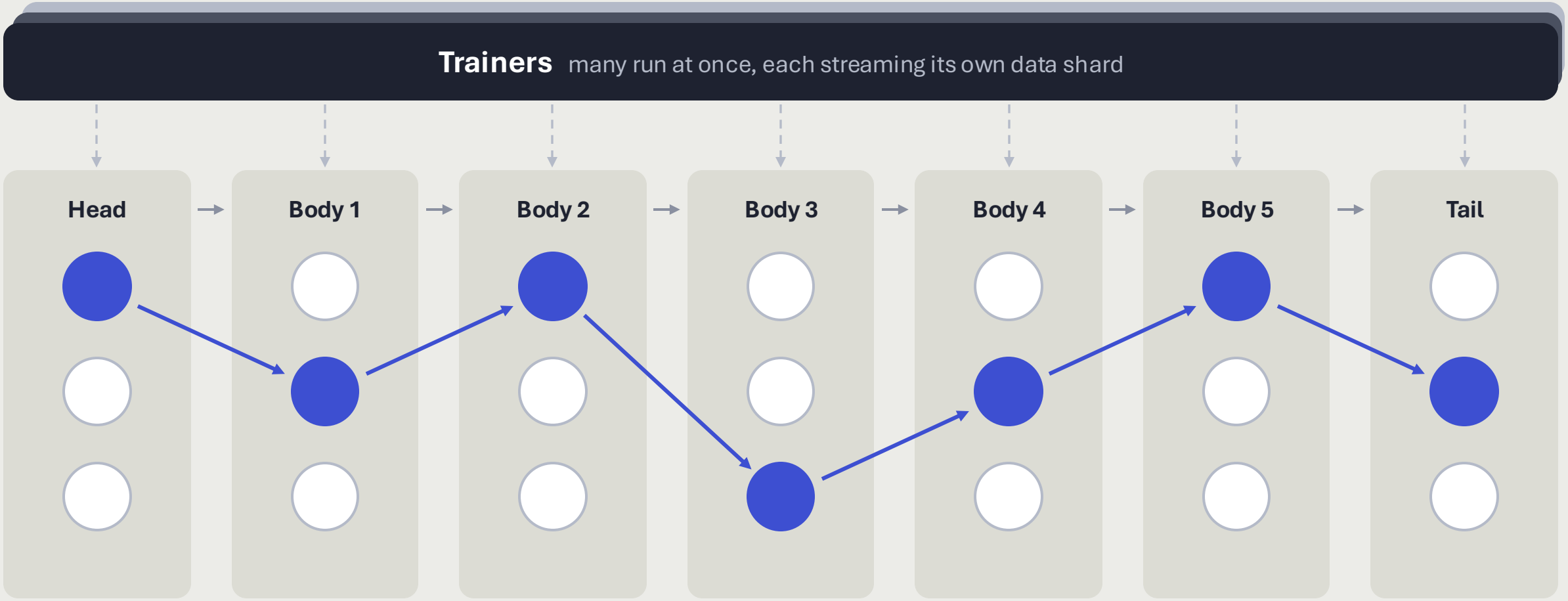
**500B**
tokens processed

**330**
contributor nodes

FineWeb-Edu · bf16 · 14 May → 23 Jun 2026

Paper-reported figures; the public dashboard shows 531.8B tokens over 51 days. Source: Agora paper §5.2–5.3; agora.pluralis.ai

# Model split into Stages




Source: Agora paper, Figure 1 and §3.1 (arxiv.org/html/2607.13332v1#S3.SS1)

# Contributers and Reducers

| | Reducer node | Contributor node |
|---|---|---|
| Averaging | **Averages one weight slice, sends it back** | **Sends its slices, applies the averages** |
| If it drops out | Its slice keeps local values that round | Round finishes without it |

Both roles are workers in the same stage. Source: Agora paper §3.3.1 (reducer role, fault tolerance), §5.1 (run setup)

# AllReduce Operation

## Hivemind

Every peer holds the full model

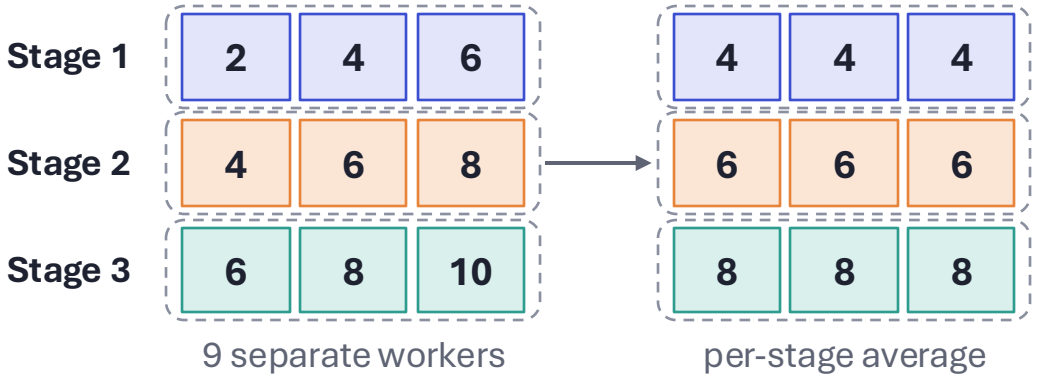



**One average over all stages, on every peer**

**Every peer averages the whole model with all peers**

## Agora

Each worker holds one stage

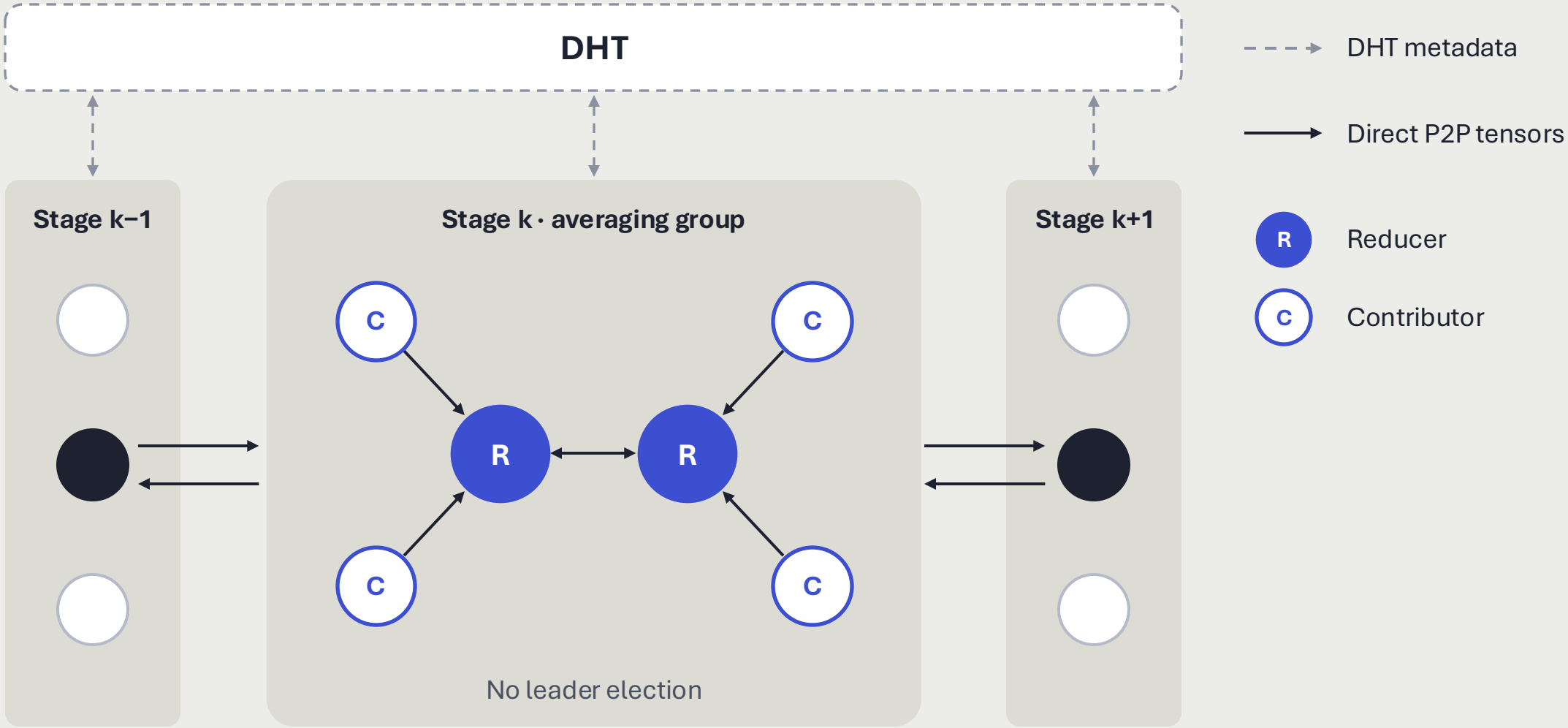



**A separate average inside each stage**

**Workers average only with replicas of their own stage**

Sources: Hivemind optimizer (hivemind/optim/optimizer.py:35); Agora paper §3.2 (arxiv.org/html/2607.13332v1#S3.SS2)

# Averaging




Source: Agora paper, Algorithm 3 and §5.9.2 (arxiv.org/html/2607.13332v1#S5.SS9.SSS2)

# Reducers average 5% of the weights every 20 steps

## When and what

### Every 20th stage step



step 1 — **step 20: average**

1 step = 2,048 sequences across the stage

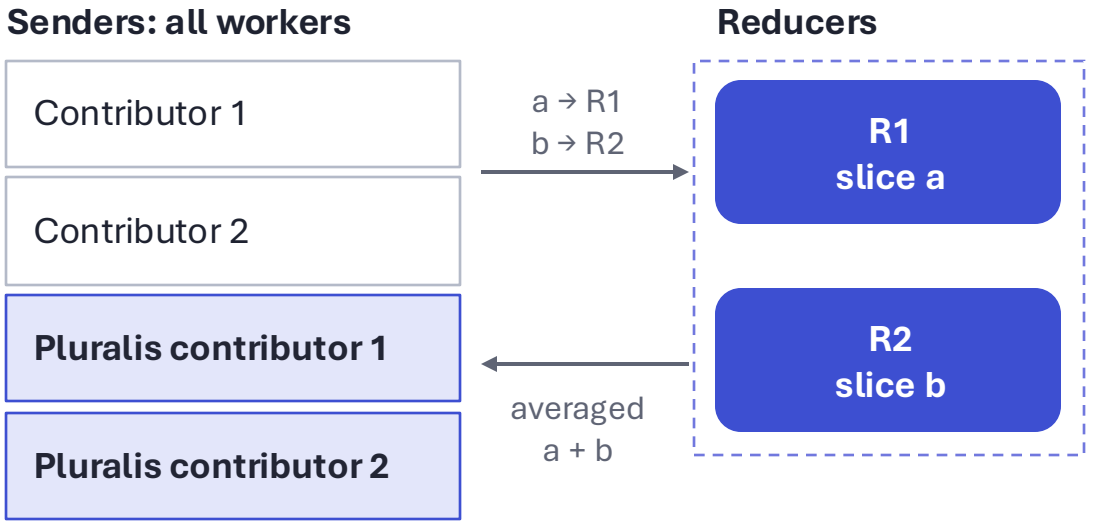
### A rotating 5% of the weights



All weights covered every 400 steps

### Training never pauses

## Who averages



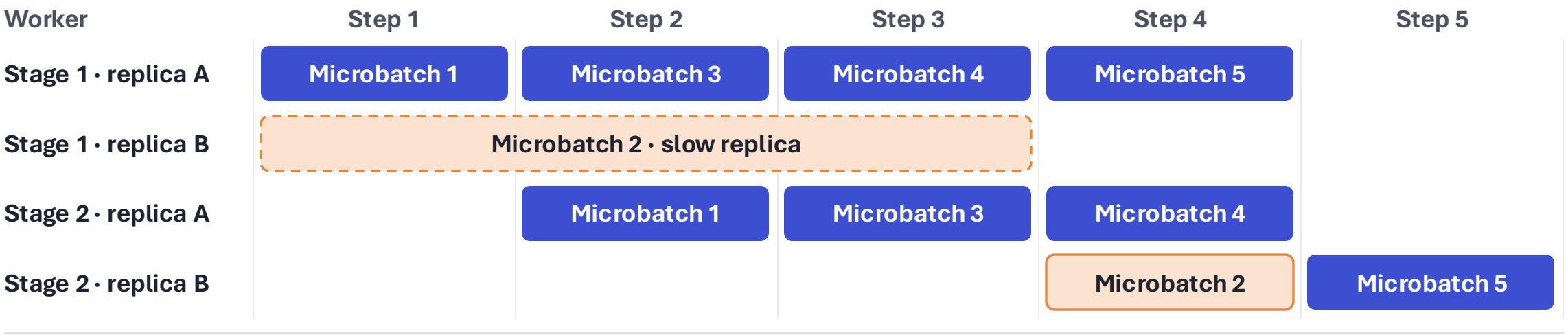

Both averaged slices return to all workers

Pluralis-8B values. Source: Agora paper §3.3.1, Algorithm 3

# A slow replica delays only its own microbatch

## Timeline with one slow replica

Stages 1 and 2, two replicas each




**Trainers route new microbatches to fast replicas**

Illustrative example. Source: Agora paper §3.2 (arxiv.org/html/2607.13332v1#S3.SS2)

# Joining happens once; averaging repeats every round

One peer's view

1
**Load config**
Model + stage

2
**Join the DHT**
Bootstrap peers

3
**Advertise the key**
prefix + step

4
**Build a roster**
No leader

5
**All-reduce**
Peer-to-peer

Once, on join

Every round: every 20 steps, new key

Source: Agora paper §3.3.1, Algorithm 3, §5.9.2; step 1 settings from our harness

# Agora is still centralized in four places

## What Pluralis kept in this run

### Admission

One Authorizer admits every node

Single gatekeeper

### Head and tail

Contributors join body stages only

Head 8 · tail 16 nodes

### Averaging

Only Pluralis nodes reduce

Contributors only send

### Starting state

Pluralis ran the first 5,000 steps

Joiners load its saves

## In the paper's words

**“This run was not decentralized and required specific Pluralis nodes.”**

Source: Agora paper §3.3 (training roles), §3.3.1 (reducer role), §3.4.3 (stage bottlenecks), §3.6.1 (Authorizer), §5.1 (run setup)

# Contributors needed fast links and large GPUs; 669 joined and 607 left over 40 days

## Admission requirements

**≥ 200 Mbps**
download
Median: 703 Mbps

**≤ 80 ms**
round-trip time
Median: 28 ms

**≥ 24 GB**
GPU memory

**≥ 80 GB**
system RAM

## Membership over the 40-day run

**669**
joins
≈ 17 per day

**607**
departures
≈ 15 per day

**63**
peak concurrent contributors
cap: 60

Rejoins count as new events. Source: Agora paper §5.2–5.3 (admission thresholds, measured contributor network, membership events)



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# Agora run results: Pluralis-8B telemetry

### Matchmaking

**5 s**

median time to form a roster

0 s — 40 s window

Typical roster: 16 peers

### DHT lookups

**67 ms**

median read (write: 68 ms)

99th percentile: 471 ms read, 1,110 ms write

Slow intervals: 0.5–1.1 s

### Weight averaging

**99.3%**

of averaged weight data received

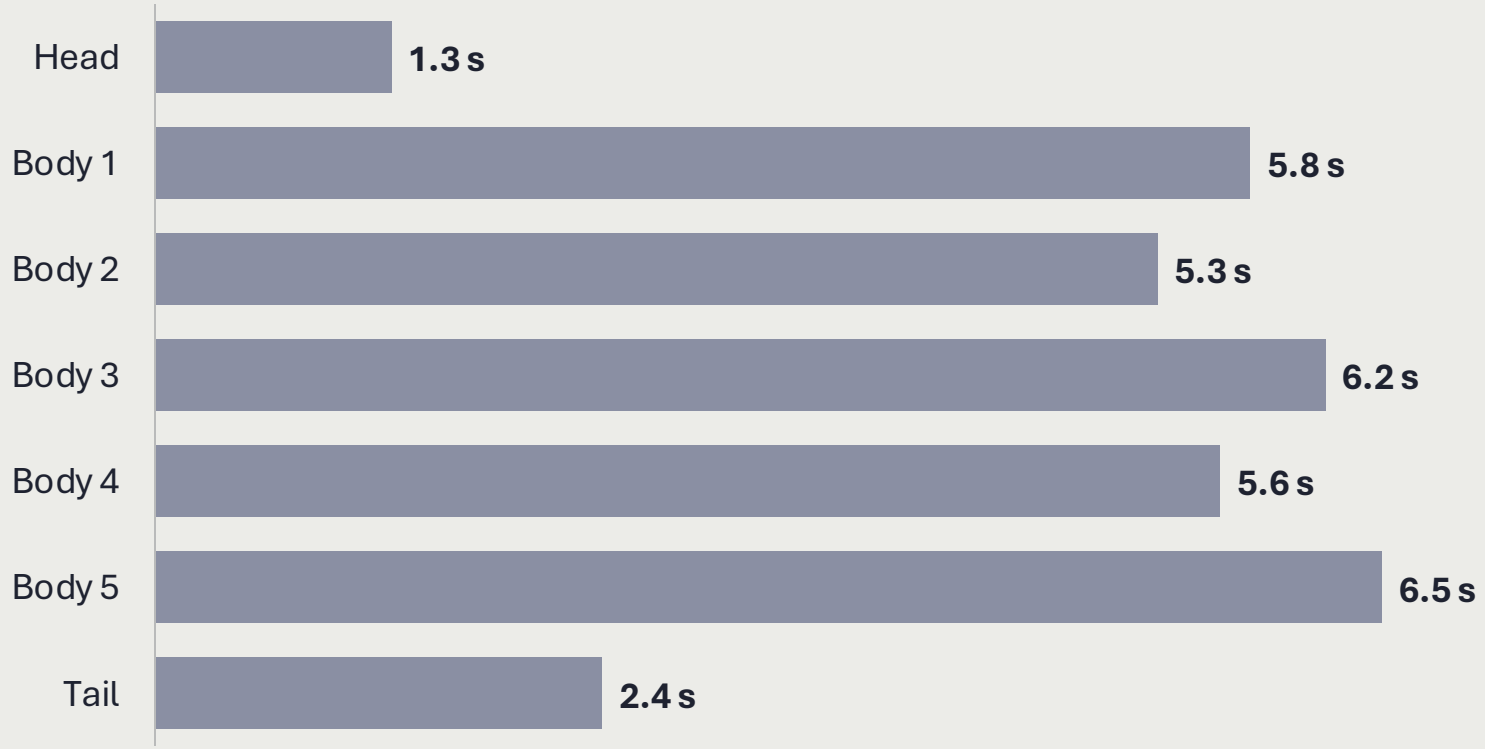
0% — 100%

Reducers: 99.7% of contributions

Medians and means of ~3-hour interval values from the live Pluralis-8B run. Source: agora.pluralis.ai telemetry API; 40 s window from the paper's config.

# Rosters formed in about 5 s, well inside the 40 s matchmaking window

**Median roster formation time by stage**




**5 s**

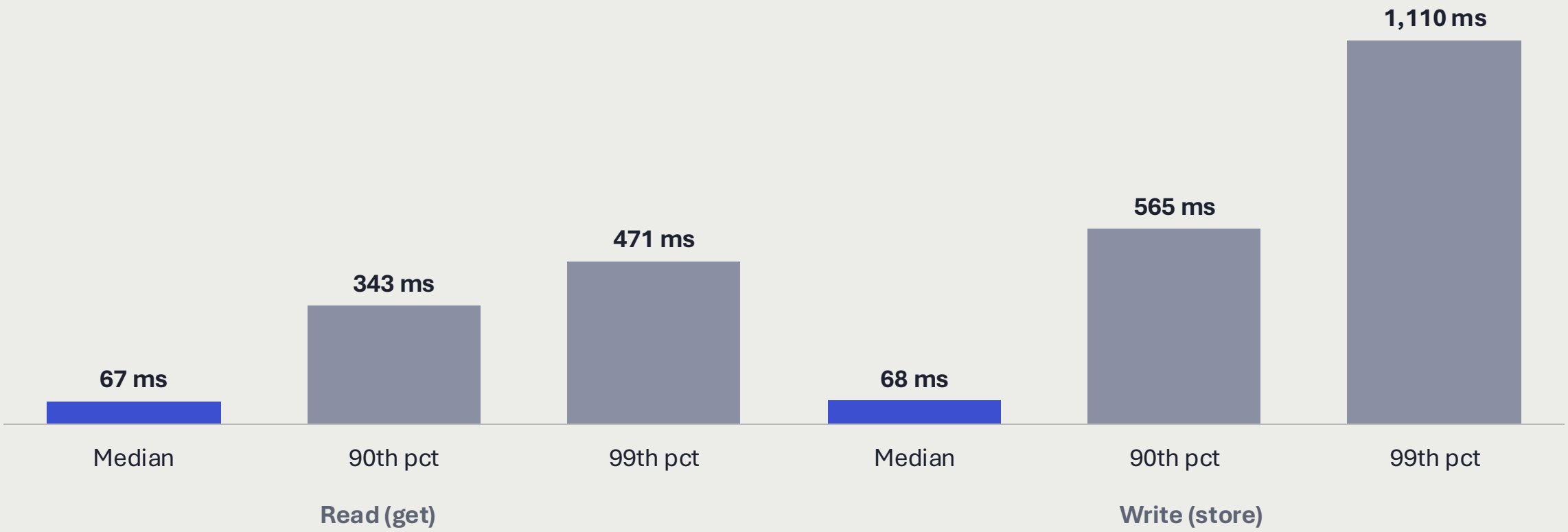
median roster time




**Typical roster: 16 peers**

Medians of ~3-hour exported interval values, not individual rounds. Source: agora.pluralis.ai telemetry API; Agora config (40 s window)

# DHT operation latency

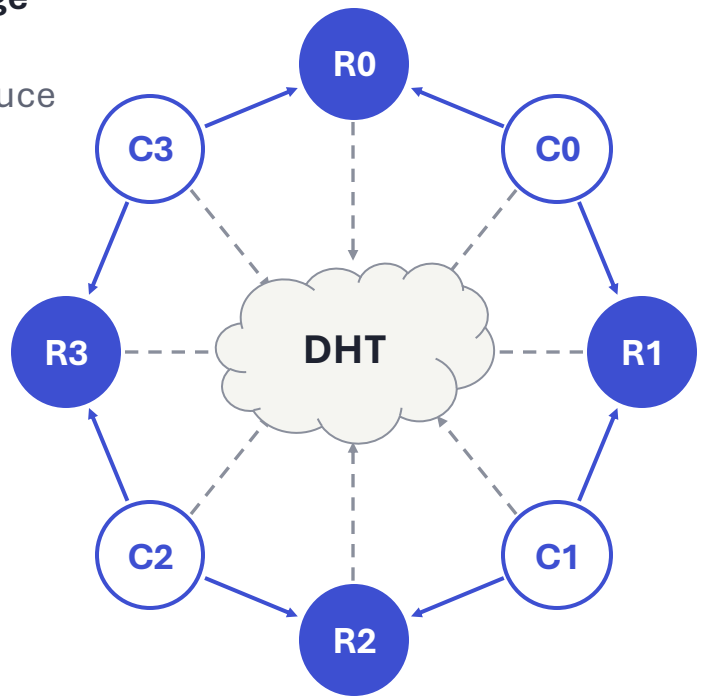



Percentiles across ~3-hour interval averages per worker, not individual calls. Source: agora.pluralis.ai telemetry API

# Simulation results

**One laptop · one stage**

Native Agora DHT, matchmaking, all-reduce




Contributor

Advertisement

Averaging

Reducer

| Result | Seed 8101 | Seed 8102 |
|---|---|---|
| **Verified rounds** | **5/5** | **5/5** |
| **Matching rosters** | **5/5** | **5/5** |
| **Ready-to-roster** | 1.087 s | 0.952 s |
| **Failed tensor parts** | **0** | **0** |
| **Mean loss** | 0.7953 → 0.00188 | 1.1724 → 0.00149 |

**Network:** injected delay $(RTT_i + RTT_j)/2$ ms per leg, RTTs from the 8B run · 40 s matchmaking · 150 s timeout

**Training:** 100-value CPU model, synthetic data · 5% of weights averaged every 20 steps · 2 s steps · 5 rounds/seed

Timing means use 40 peer-round observations per seed. Source: baseline8b_pilot_20261013_a trial configuration and verified report